Proof. Let $\sigma_1, \ldots, \sigma_n$ be the singular values of H_ℓ. Given these singular values, one can compute $V(H_\ell)$ as

$$(V(H_\ell))^2 = \sum_{i=1}^{n} \left(\sigma_i^2 - \frac{1}{n}\right)^2.$$

According to Eq. (10), $\sum_{i=1}^{2} \sigma_i^2 = 1$ holds. The proof is an immediate consequence of this propery.

$$\begin{aligned} V^2(H_\ell) &= \sum_{i=1}^{n} \sigma_i^4 - 2 \underbrace{\left(\sum_{i=1}^{n} \sigma_i^2\right)}_{=1} \frac{1}{n} + \frac{1}{n} \\ &= \sum_{i=1}^{n} \sigma_i^4 - \frac{1}{n}. \end{aligned}$$

Fixing σ_n, the maximum of $\sum_{i=1}^{n} \sigma_i^4 - \frac{1}{n}$ subject to $\sum_{i=1}^{n} \sigma_i^2 = 1$ is met when $\sigma_1^2 = 1 - (n-1)\sigma_n^2$ and $\sigma_2 = \cdots = \sigma_n$. Therefore,

$$V^2(H_\ell) \leq 2(n-1)^2 \underbrace{\left(\frac{1}{n} - \sigma_n^2\right)^2}_{=\widehat{V}^2(H_\ell(X))}$$

holds true. Taking the square root of both sides concludes the proof. □

Appendix B. Proof of Theorem 1

The proof of Theorem 1 relies on the following Theorem that characterizes the change of $\widehat{V}$ in consecutive layers.

Theorem 5. *Sequence $\{H_\ell\}$ obeys*

$$\mathbb{E}\left[\widehat{V}(H_{\ell+1}) | H_\ell\right] \leq \left(1 - \frac{2}{3}\left(\frac{1}{n} - \widehat{V}(H_\ell)\right)\right) \widehat{V}(H_\ell) + \frac{1}{\sqrt{d}}.$$

Notably, the above result does not rely on Assumption $\mathcal{A}_1$. Assuming that Assumption $\mathcal{A}_1$ holds, we complete the proof of Theorem 1. Combining the last Theorem by this assumption, we get

$$\mathbb{E}\left[\widehat{V}(H_{\ell+1})\right] \leq \left(1 - \frac{2}{3}(\alpha)\right) \mathbb{E}\left[\widehat{V}(H_\ell)\right] + \frac{1}{\sqrt{d}}$$

Induction over ℓ yields

$$\begin{aligned} \mathbb{E}\left[\widehat{V}(H_{\ell+1}(X))\right] &\leq \left(1 - \frac{2}{3}\alpha\right)^\ell \mathbb{E}\left[\widehat{V}(H_1)\right] + \left(\sum_{k=1}^{\ell} (1 - \frac{2}{3}\alpha)^k\right) \frac{1}{\sqrt{d}} \\ &\leq \left(1 - \frac{2}{3}\alpha\right)^\ell \mathbb{E}\left[\widehat{V}(H_1)\right] + \frac{3}{2\alpha\sqrt{d}} \end{aligned}$$

An application of Lemma 4 completes the proof:

$$\begin{aligned} \mathbb{E}\left[V(H_{\ell+1})\right] &\leq 2n\mathbb{E}\left[\widehat{V}(H_{\ell+1})\right] \\ &\leq 2\left(1 - \frac{2}{3}\alpha\right)^\ell + \frac{3n}{2\alpha\sqrt{d}} \end{aligned}$$

B.1. **Stability analysis without $\mathcal{A}_1$.** Using Theorem 5, we can prove stability of the chain $\{H_\ell\}$ without Assumption $\mathcal{A}_1$. After rearrangement of terms in Theorem 5, we get

$$\mathbb{E}\left[\widehat{V}(H_{\ell+1})|H_\ell\right] - \widehat{V}(H_\ell) \leq -\frac{2}{3}\widehat{V}(H_\ell)\left(\frac{1}{n} - \widehat{V}(H_\ell)\right) + \frac{}{\sqrt{d}}$$

Taking the expectation over H_ℓ and average over ℓ yields

$$\mathbb{E}\left[\frac{1}{\ell}\sum_{k=1}^{\ell}\widehat{V}(H_k)\left(\frac{1}{n} - \widehat{V}(H_k)\right)\right] \leq \left(\frac{3\mathbb{E}\left[\widehat{V}(H_0)\right]}{2\ell}\right) + \frac{3}{2\sqrt{d}}$$

Appendix C. Proof of Theorem 5

C.1. **Spectral decomposition.** Consider the SVD decomposition of H_ℓ as $H_\ell = U\text{diag}(\sigma)V^\top$ where U and V are orthogonal matrices. Given this decomposition, we get

$$W_\ell H_\ell = \underbrace{W_\ell U}_{W}\,\text{diag}(\sigma)V^\top \tag{12}$$

Since W_ℓ is Gaussian and U is orthogonal, entities of matrix W are also i.i.d. standard normal. This decomposition will be repeatedly used.

C.2. **Concentration.** Consider matrix $C_{\ell+1} := H_{\ell+1}^\top H_{\ell+1}$ whose eigenvalues are $\sigma_1^2, \ldots, \sigma_n^2$. The SVD decomposition of H_ℓ allows writing $C_{\ell+1}$ as the average of vectors

$$C_{\ell+1} = \frac{1}{d}\sum_{i=1}^{d}\left(\frac{w_i \odot \sigma}{\|w_i \odot \sigma\|_2}\right)^{\otimes 2}$$

where $w_i \in \mathbb{R}^n$ are rows of matrix W in Eq.(12) and $\sigma \in \mathbb{R}^n$ is the vector of singular values of H_ℓ. Thus, conditioned on σ, $C_{\ell+1}$ is an empirical average of i.i.d. random vectors. This allows us to prove that this empirical average is concentrated around its expectation. The next lemma states this concentration.

Lemma 6. *The following concentration always holds*

$$\mathbb{E}_{W_\ell}\|C_{\ell+1} - \mathbb{E}_{W_\ell}\left[C_{\ell+1}\right]\|^2 \leq 1/d$$

where

$$\mathbb{E}_{W_\ell}\left[C_{\ell+1}\right] = \textit{diag}(p_1(\sigma), \ldots, p_n(\sigma)), \quad p_i(\sigma) := \mathbb{E}\left[\frac{\sigma_n^2 w_n^2}{\sum_{k=1}^n \sigma_k^2 w_k^2}\right], \quad w_i \overset{i.i.d.}{\sim} \mathcal{N}(0,1) \tag{13}$$

The concentration of $C_{\ell+1}$ allows us to prove that the Lyapunov function $\widehat{V}(H_{\ell+1})$ is concentrated around $1/n - p_n(\sigma)$.

Lemma 7. *The following holds*

$$\mathbb{E}_{W_\ell}\left[\left(\widehat{V}(H_{\ell+1}) - (1/n - p_n(\sigma))\right)^2\right] \leq \frac{1}{d}$$

.

The last lemma allows us to predict the value of random variable $\widehat{V}(H_{\ell+1})$ by deterministic term $1/n - p_n(\sigma)$.

C.3. **Contraction.** The decay in $\widehat{V}(H_{\ell+1})$ with ℓ is due to term $1/n - p_n(\sigma)$ in the last lemma. This term is less than (or equal to) $V(H_\ell)$.

Lemma 8. *For $p_n(\sigma)$ defined in Eq. (13), the following holds:*

$$\left(\frac{1}{n} - p_n(\sigma)\right) \leq \left(1 - \frac{2}{3}\left(\frac{1}{n} - \widehat{V}(H_\ell)\right)\right)\widehat{V}(H_\ell).$$

Combining the last lemma by Lemma 7 concludes the proof of Theorem 5:

$$\mathbb{E}_{W_\ell}\left[\widehat{V}(H_{\ell+1})\right] \leq \left(1 - \frac{2}{3}\left(\frac{1}{n} - V(H_\ell)\right)\right)\widehat{V}(H_\ell) + \frac{1}{\sqrt{d}}. \tag{14}$$

To complete the proof, we prove Lemmas 6, 7, and 8.

C.4. **Proof of Lemma 6.** Given the spectral decomposition of H_ℓ in Eq. (12), we compute element ij of $C_{\ell+1}$, which is denoted by $[C_{\ell+1}]_{ij}$:

$$[C_{\ell+1}]_{ij} = [A^\top A]_{ij} = \frac{1}{d}\sum_{k=1}^{d} A_{ki}A_{kj}, \quad A_{ki} = W_{ki}\sigma_i/\sqrt{v_k}$$

where $v_k = \sum_{m=1}^{n} W_{km}^2\sigma_m^2$. Since W_{km} are zero mean and unit variace, we get

$$\begin{aligned}
\mathbb{E}[C_{\ell+1}]_{ij} &= 0 \\
\mathbb{E}[C_{\ell+1}]_{ij}^2 &= \frac{1}{d^2}\sum_{k=1}^{d} A_{ki}^2A_{kj}^2 + \frac{1}{d^2}\sum_{k,k'}^{d} \underbrace{\mathbb{E}\left[A_{ki}^2A_{kj}^2A_{k'i}^2A_{k'j}^2\right]}_{=0} \\
&= \frac{1}{d^2}\sum_{k=1}^{d} A_{ki}^2A_{kj}^2
\end{aligned}$$

holds for $i \neq j$. By summing up over $i \neq j$, we get

$$\sum_{i\neq j}\mathbb{E}[C_{\ell+1}]_{ij} = \frac{1}{d^2}\left(\sum_k\left(\sum_i A_{ki}^2\right)^2 - \sum_{ik} A_{ki}^4\right)$$

For the diagonal elements, we get

$$\begin{aligned}
\mathbb{E}\left[C_{\ell+1}\right]_{ii} &= \frac{1}{d}\sum_{k=1}^{d}\mathbb{E}A_{ki}^2 \\
&= \frac{1}{d}\sum_{k=1}^{d}\mathbb{E}A_{ki}^2 \\
&= \underbrace{\frac{1}{d}\sum_{k=1}^{d}\mathbb{E}\left[\frac{W_{ki}^2\sigma_i^2}{\sum_{j=1}^{n}W_{kj}^2\sigma_j^2}\right]}_{p_i(\sigma)}
\end{aligned}$$